

value to i . For example, in the sequence, -3, 4, 3, 34, 10, we have $a_3=3$, which satisfies the criterion; whereas in the sequence -3, 40, 2, 12, 6, there is no such index i . What is the complexity of your algorithm?

- (10) Let's suppose you are given an array A of N sorted numbers that has been circularly shifted k positions to the right. For example, (34, 40, 2, 12, 28) is a sorted array that has been circularly shifted $k=2$ positions, while (34, 40, 45, 61, 2, 12, 28) has been shifted $k=4$ positions. Suppose you know what k is, give an $O(1)$ algorithm to find the largest number in A . If you do not know the value of k , can you give an algorithm with a complexity of $O(N)$ for finding the largest number in A ? Can you improve your solution to an algorithm of complexity $O(\log N)$?
- (11) Design a data structure that allows you to search, insert and delete an integer X in $O(1)$ time in a table (i.e., constant time, independent of the total number of integers stored). Assume that $1 \leq X \leq N$. Also assume that the maximum number of integers in the table can only be M at any one time. You are given $M+N$ units of space available for the table. Though at first you may want to use a flags array of M flags, note that you cannot initialise the flags array since that would take a time of $O(M)$ and hence is not permitted. So you need to work with a flags array that can contain uninitialised data.
- (12) You are given an array of N integers. Some of the integers are positive and others negative. You have to partition the array such that all the negative elements occur first and then all the positive elements. (There is no need to sort the elements within the sub-partition of all negative and all positive elements.) What is the time complexity of your solution?
- (13) Among all the spanning trees of a weighted and connected graph, the one (possibly more) with the least total weight is called a minimum spanning tree (MST). There are well-known algorithms for determining the MST of a given weighted connected graph G such as Prim's and Kruskal's algorithm. Our question is as follows: Given a minimum spanning tree, is the path between a pair of vertices in a minimum spanning tree necessarily the shortest path between the two vertices in the full graph G ? Give a proof or a counter example.

Graphs and trees

- (14) All of you are familiar with tree traversals. Pre-order, in-order, and post-order traversals are the three types of tree traversals. If the root node is visited before either of its sub trees, this is a pre-order. If the root node is visited after visiting the left sub tree and right sub tree, then it is a post-order traversal. If the left sub tree is visited first, then the root node, followed by the right sub tree, then it is an in-order traversal. Given only the pre-order and in-order traversals of a tree, is it possible to construct a unique binary tree from it?
- (15) Given an undirected graph $G(V, E)$, where V is the set of vertices and E is the set of edges, and given that all

vertices have an even degree, can you give an algorithm to determine whether there exists a subset S of k vertices in a graph G , such that every edge in G is incident upon at least one vertex in S ?

A few more challenging questions

- (16) Given three arrays A , B and C of integers containing positive and negative numbers, we have to determine if there are three elements $a \in A$, $b \in B$ and $c \in C$ such that their sum $a+b+c$ is equal to zero.
- (17) Given an input array of distinct 100,000 integers to be sorted, you are told that you have memory available only for 1,000 numbers at a time. You have no constraints in terms of disk storage. Can you come up with an algorithm to sort the 100,000 numbers? If you are told that the range of values of the input array is also 1 to 100,000, how would your solution change? What would be your solution if the input array contained duplicates?
- (18) Given the following code snippet:

```
bool foo (int n)
{
    for (i=3; i <= sqrt(n)+0.5; i+=2)
    {
        if (n % i == 0)
        {
            return false;
        }
    }
    return true;
}
```

can you figure out what is the purpose of the function *foo*?

- (19) Given a 32-bit integer N , come up with an algorithm to find the number of zeroes in the binary bit representation of N . What is the time complexity of your algorithm?
- (20) Let S be a sorted array of n integers. Give an algorithm that finds the pair of elements $x, y \in S$ that maximises $|x-y|$. What is the time complexity of your solution?

Readers are requested to send their solutions to these questions, which I will discuss in next month's column. Also, if you are preparing for an interview in the computer science programming field, you may find it useful to visit my programming interviews discussion group 'Computer Science Interview Training (India)' on LinkedIn (www.linkedin.com).

If you have any favourite programming puzzles that you would like to discuss on this forum, please send them to me, along with general feedback at sandyaswam_AT_yahoo_DOT_com. Till we meet again next month, happy programming. 

About the author:

Sandya Mannarswamy, The author is a specialist in compiler optimisation and works at Hewlett-Packard India. She has a number of publications and patents to her credit, and her areas of interest include virtualisation technologies and software development tools.